EXPERIMENT 12

Earley Parser for Context-Free Grammars

AIM

To implement an Earley chart parser for Context-Free Grammars in Python capable of handling ambiguous and recursive grammatical rules.

PROCEDURE

1. Define a Context-Free Grammar with potential syntactic ambiguity (e.g. PP attachment).
2. Initialize Earley Parser utilizing chart data structures with Predictor, Scanner, and Completer states.
3. Tokenize an ambiguous sentence ('I saw the man with a telescope').
4. Execute the chart parser to build states across positions 0 to n.
5. Extract and print all valid parse trees resulting from the chart parsing process.

PROGRAM

```
import nltk
from nltk import CFG
from nltk.parse.earleychart import EarleyChartParser

ambiguous_grammar = CFG.fromstring("""
    S -> NP VP
    PP -> P NP
    NP -> Det N | NP PP | 'I'
    VP -> V NP | VP PP
    Det -> 'a' | 'the'
    N -> 'man' | 'telescope' | 'park'
    V -> 'saw'
    P -> 'with' | 'in'
""")

parser = EarleyChartParser(ambiguous_grammar)

sentence = "I saw the man with a telescope".split()
print(f"Input Sentence: {' '.join(sentence)}\n")

trees = list(parser.parse(sentence))
print(f"Total Parse Trees Generated by Earley Parser: {len(trees)}\n")

for i, tree in enumerate(trees, 1):
    print(f"--- Parse Tree {i} ---")
    tree.pretty_print()
```

OUTPUT

```
> python3 experiment_12.py
```

```
Input Sentence: I saw the man with a telescope

Total Parse Trees Generated by Earley Parser: 2

--- Parse Tree 1 ---
            S
   _________|____
  |              VP
  |     _________|_____
  |    |         |     PP
  |    |         |    _|_
  NP   V         NP  P   NP
  |    |    _____|   |   |____
  I   saw Det    N  with Det   N
           |     |        |    |
          the   man       a telescope

--- Parse Tree 2 ---
            S
   _________|____
  |              VP
  |    __________|_
  |   |            NP
  |   |     _______|____
  |   |    |            PP
  |   |    |        ____|___
  NP  V    NP      P        NP
  |   |   _|_      |     ___|____
  I  saw Det N    with  Det      N
          |  |           |       |
         the man         a   telescope

[Process completed with exit code 0]
```

RESULT

Thus, the Earley parser was successfully implemented, correctly discovering multiple valid derivations for syntactically ambiguous context-free grammars.